

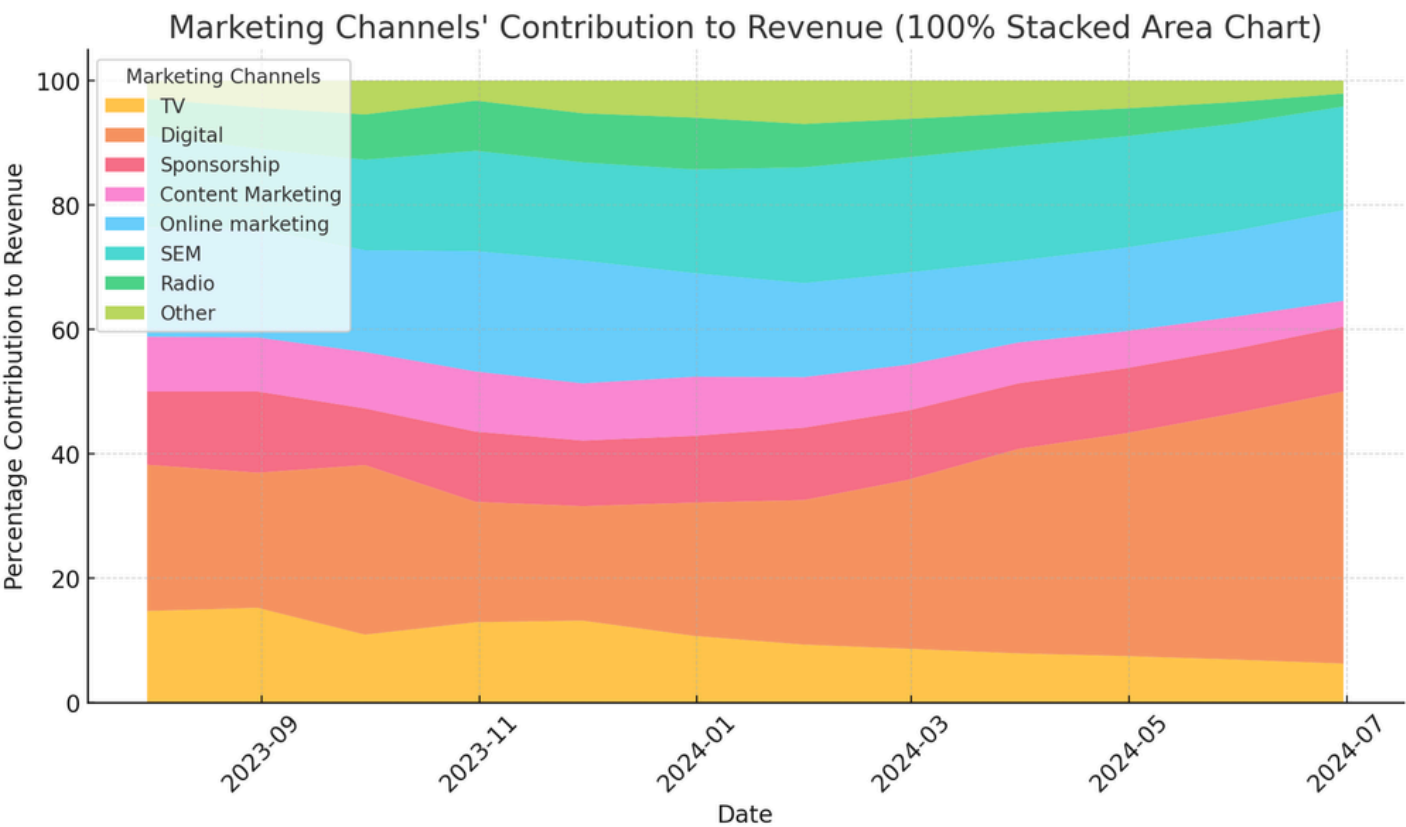
Team 2025114

Business Problem

Why Budget Optimization Matters?

ElectroMart has been overspending on inefficient marketing channels, requiring a data-driven approach to reallocate budgets and maximize ROI.

Over the last 12 months, the marketing budget has not yielded the expected revenue impact. The CFO believes they must either cut the budget or reallocate it strategically to maximize efficiency. Marketing teams need a data-driven strategy to ensure better revenue response from future marketing investments.



Objectives

Performance driver analysis

Which KPIs drive the top-line (revenue) performance?

Impact analysis on marketing ROI

What is the quantitative impact of each commercial lever on revenue?

Optimizing marketing spending

Best allocating the marketing budget for highest outcome

Marketing Mix Modeling (MMM) measures marketing impact on sales, identifies high-ROI channels, and optimizes budget allocation.

Dataset Name	Description	Key Variables Used
Customer Orders Data	Contains details of past purchases, revenue, and product categories.	Order ID, Revenue, Product Type, Discount %
Media Spend Data	Marketing budget allocation across different channels.	TV Spend, Digital Ads Spend, Social Media Spend, Print Media, Sponsorships
Sales Calendar Data	Promotional events, holidays, and seasonal effects.	Black Friday, Cyber Monday, Holiday Sales
NPS & Customer Satisfaction	Net Promoter Score & stock index for brand perception analysis.	NPS Score, Customer Retention, Stock Index
Weather Data	Regional climate trends that impact sales.	Temperature, Rainfall, Snowfall
Holiday Calendar Data	National & festive holidays affecting sales trends.	Public Holidays, Payday Effect
Product Hierarchy Data	SKU details and category classification.	Super-category, Sub-category, SKU Mapping

Data cleaning and pre-processing

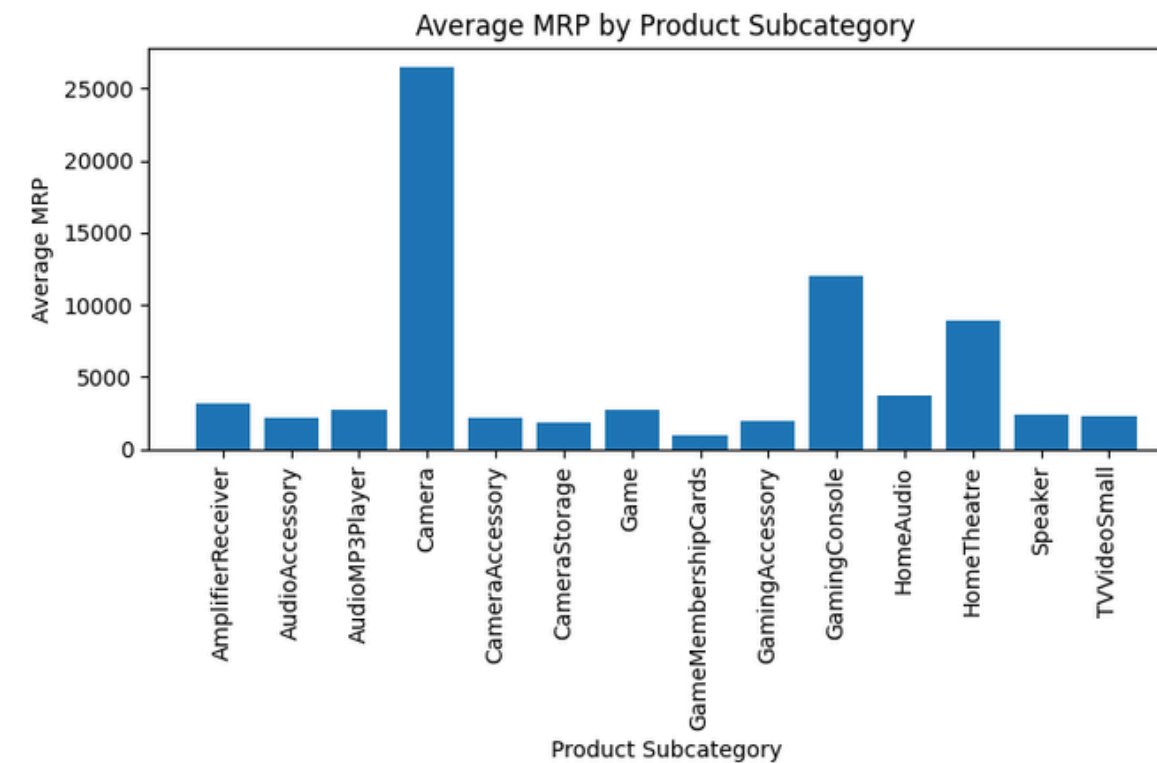
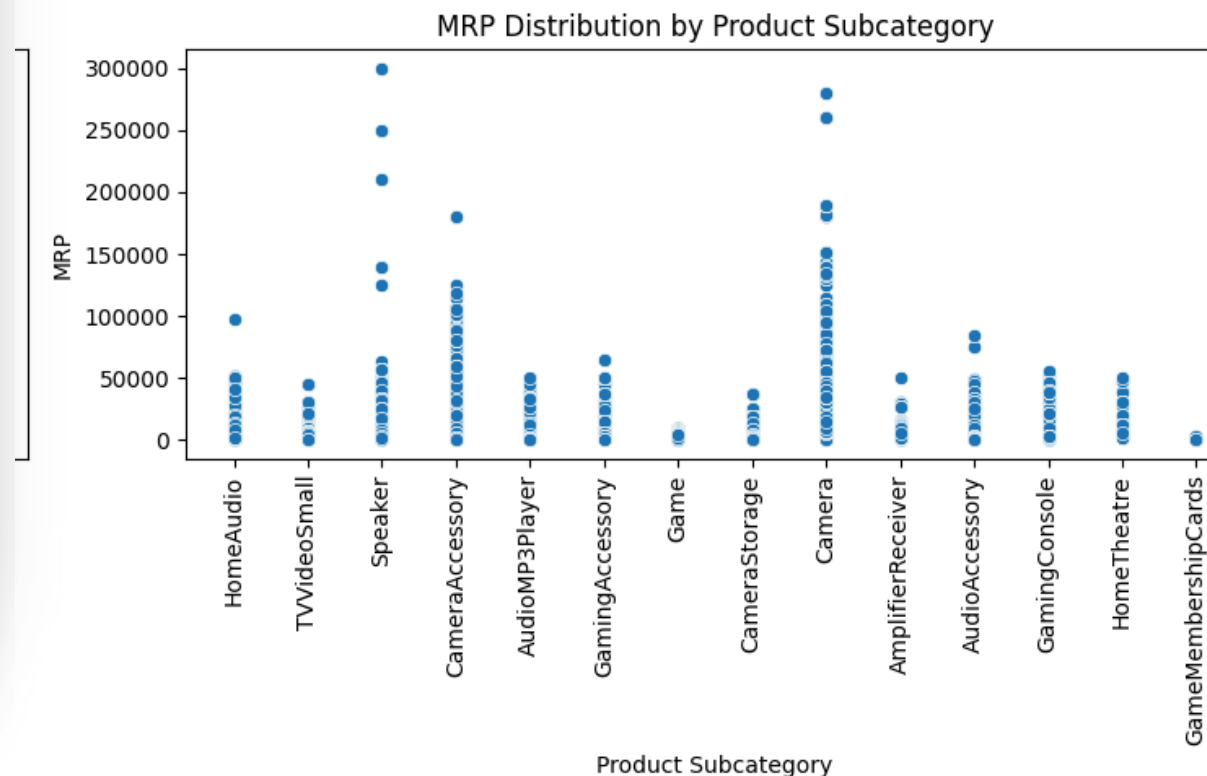
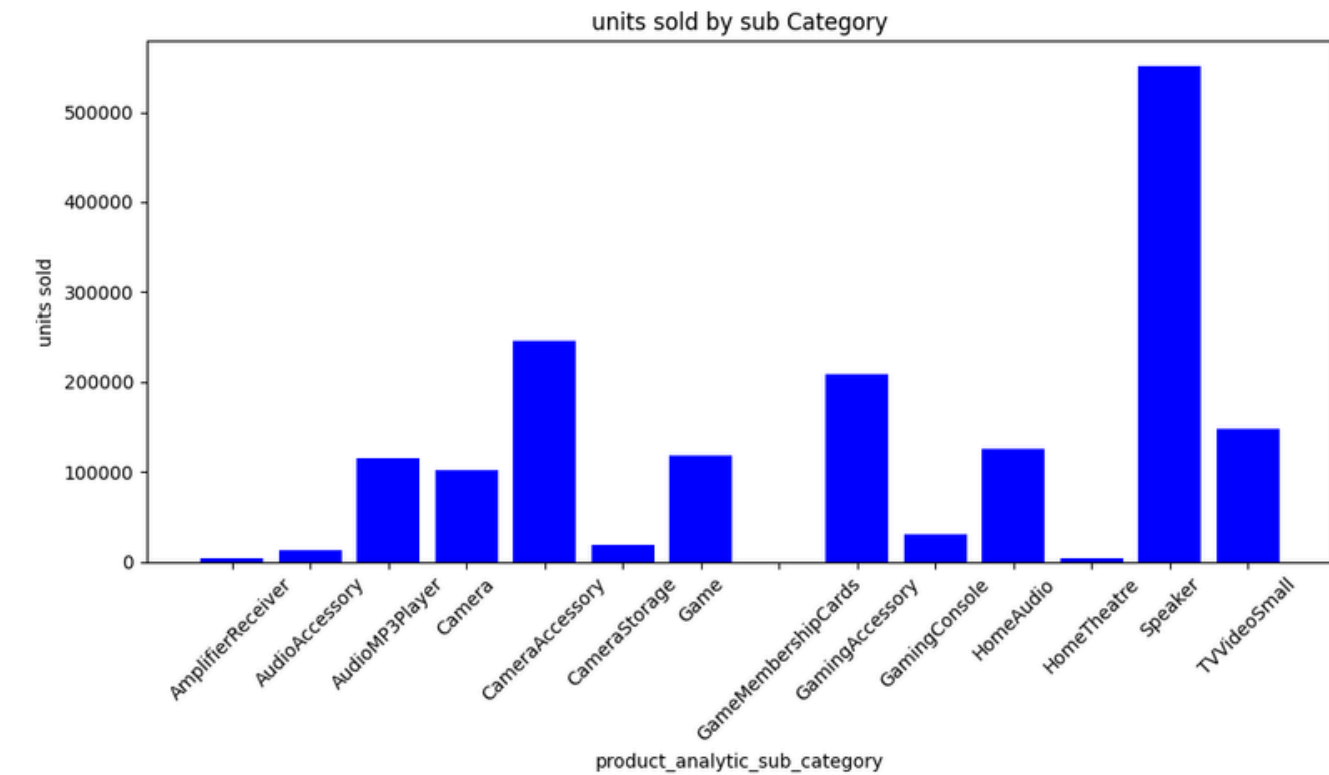
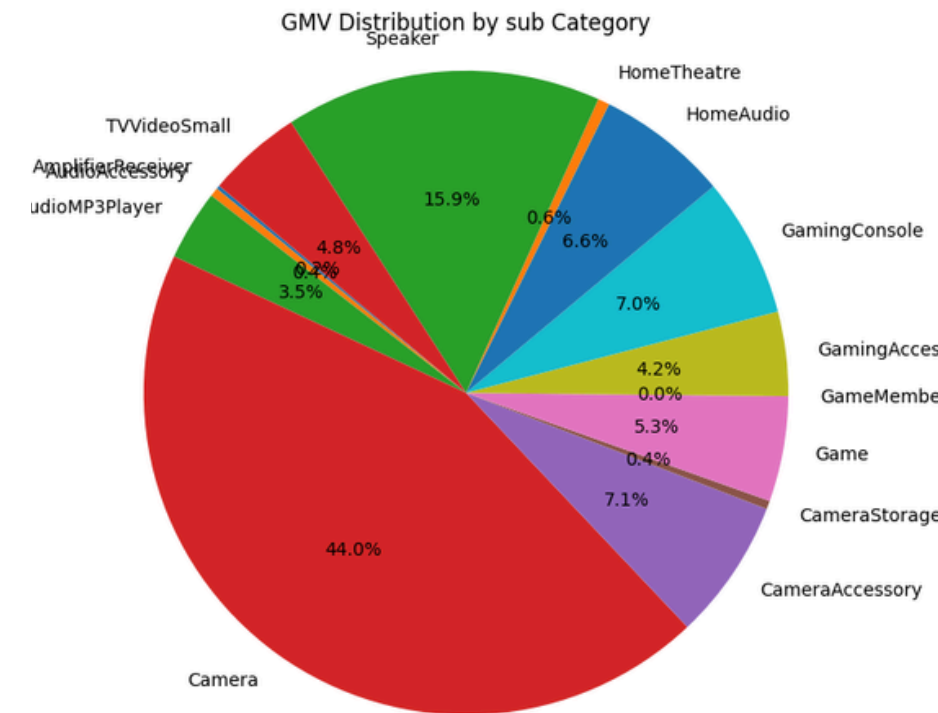
Data cleaning involved handling missing values by setting nulls to 0, interpolating weather data, and refining datasets like Stock Index, NPS, and Media Spend. Further processing included aggregating GMV monthly, merging customer orders with investment data, and computing ROI. Feature engineering classified products into luxury or mass market (based on GMV percentile) and calculated Customer Acquisition Cost (CAC) as $\text{Total Spend} / \text{New Customers}$ to assess marketing efficiency.

Dashboards

Exploratory Data Analysis

GMV distribution vs units sold

Cameras dominate GMV (44%), indicating they are high-value, low-volume products, whereas Speakers lead in sales volume but generate lower revenue. Implement premium pricing and exclusive marketing for Cameras to maximize profit, while leveraging bulk discounts and strategic bundling for high-sales categories like Speakers to drive overall revenue growth.



MRP Distribution

Cameras, Speakers, and Camera Accessories have a wide price range, catering to both budget and premium customers—ElectroMart should segment its marketing strategy, promoting premium models through high-end branding and financing options while using discounts and bundles to push budget-friendly products.

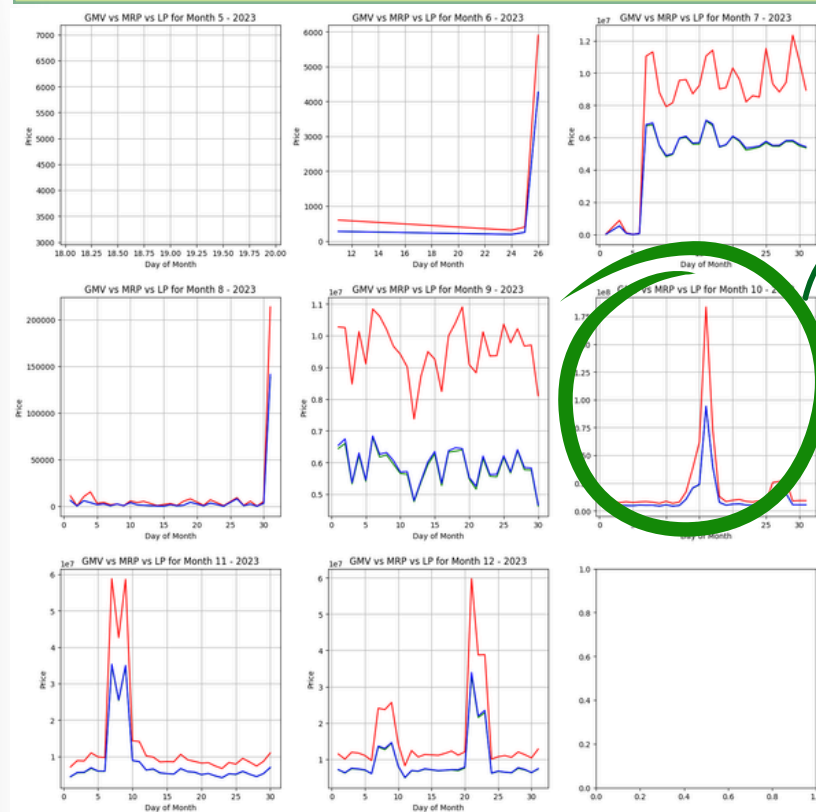
Data-Driven Insights

What is ElectroMart doing?

Linking Stock Index & Customer Behavior

Market Trends

GMV vs MRP vs LP month wise for 2023



Best Month - October 2023

Mid-Month Sales Peak: A sharp spike in GMV, MRP, and LP suggests a major promotional event driving high sales.

Discount-Driven Growth: A widened gap between MRP and LP indicates heavy discounting played a key role in boosting revenue.

Post-Sale Decline: Sales dropped after the peak, highlighting the need for strategies to maintain demand beyond promotions.

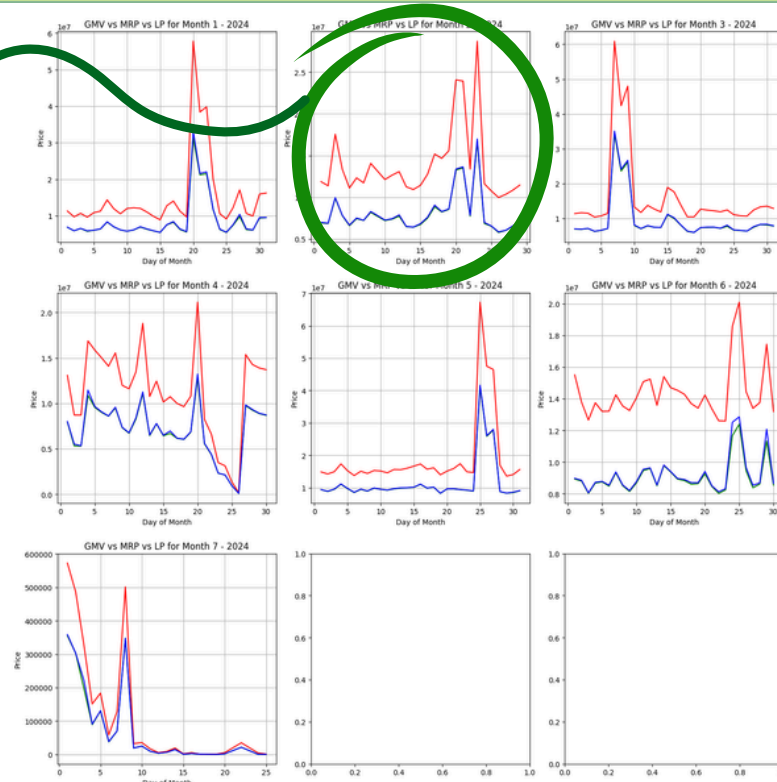
GMV vs MRP vs LP month wise for 2024

Best Month - February 2024

Major Sales Surge: A sharp peak in GMV, MRP, and LP around mid-to-late February indicates a successful promotional event.

High Discount Impact: The large gap between MRP and LP suggests that deep discounts played a crucial role in driving sales.

Post-Promotion Drop: Sales decline after the peak, highlighting the need for sustained marketing efforts to maintain demand beyond sales periods.

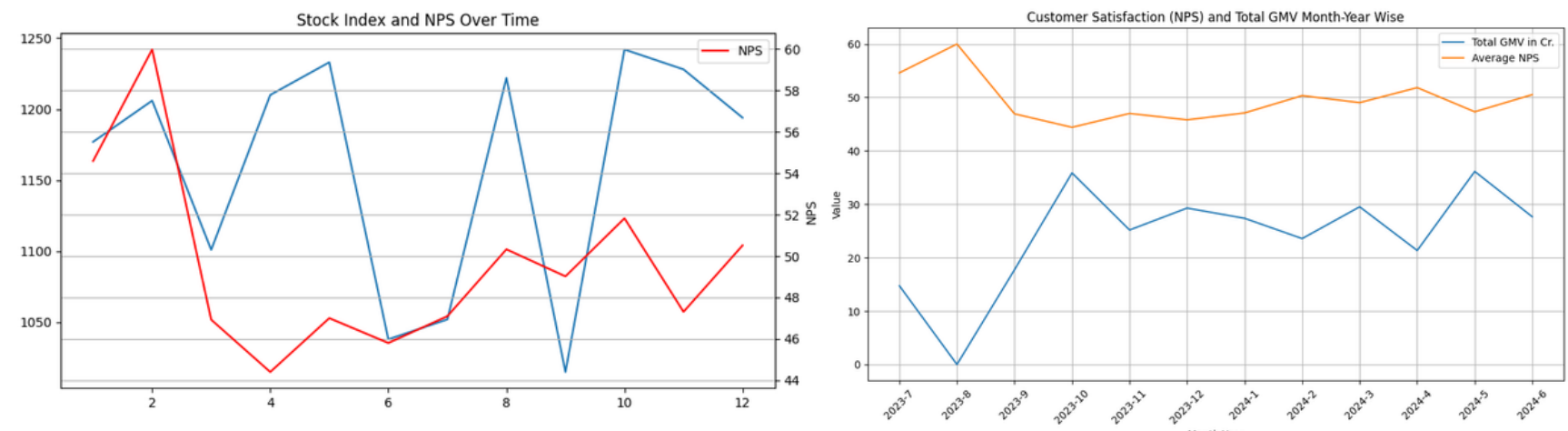


Divergence

NPS and the stock index sometimes diverge, indicating that shifts in customer sentiment don't always align with market performance, as both can be influenced by different factors.

Potential Correlation

At times, both metrics move together, suggesting a potential link where higher customer satisfaction may drive better market performance, or market success could boost customer sentiment.



NPS Stability vs. GMV Fluctuations: NPS remains stable despite GMV variations, indicating customer loyalty isn't impacted by short-term revenue dips.

Lag Effect in Customer Impact: NPS growth often precedes GMV growth, suggesting a delayed impact of customer satisfaction on revenue.

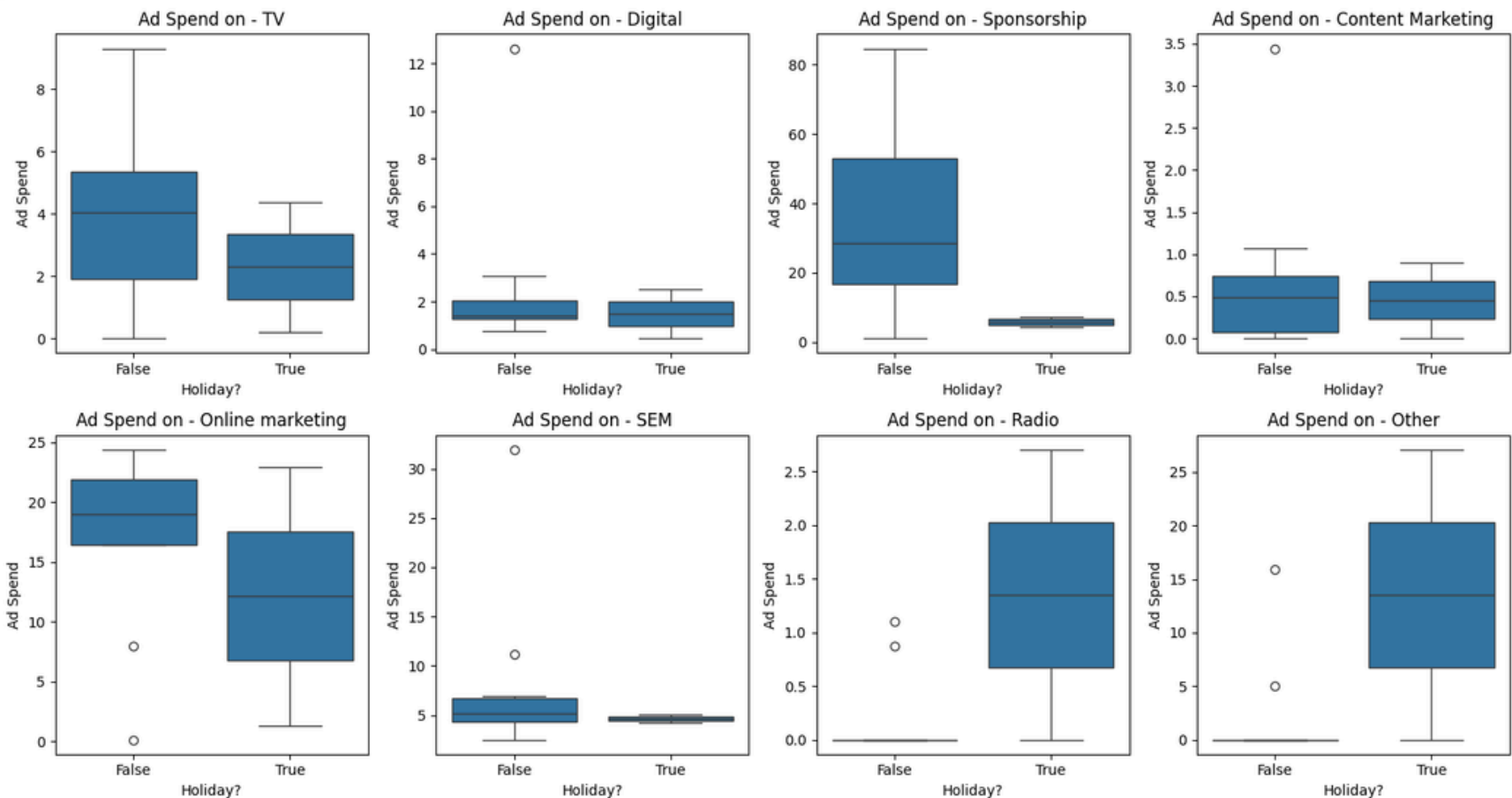
Leveraging High-NPS Customers: Engage satisfied customers with exclusive offers during slow periods to convert loyalty into repeat purchases.

Long-Term Revenue Strategy: Investing in customer experience improvements today can lead to sustained GMV growth in the future.

Influence of Holidays & Payment Trends

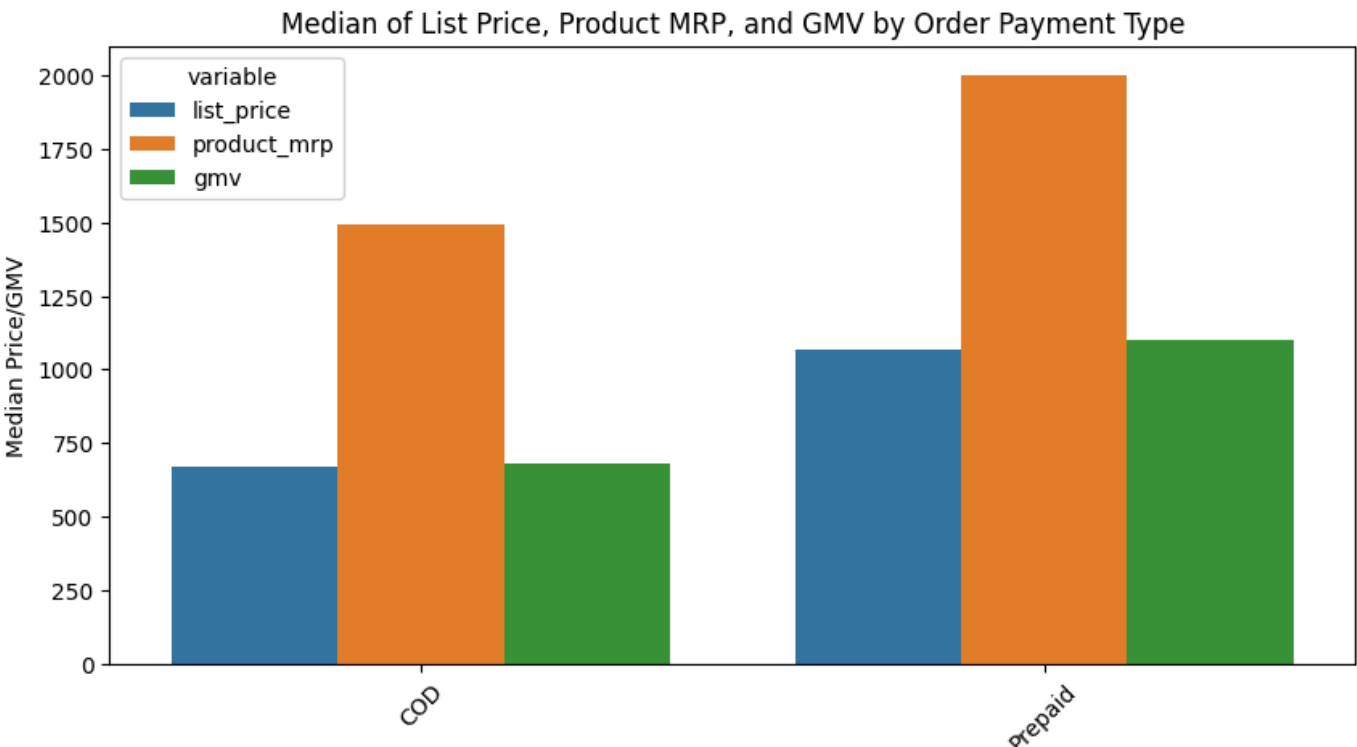
Insights

Holidays vs Non-Holidays

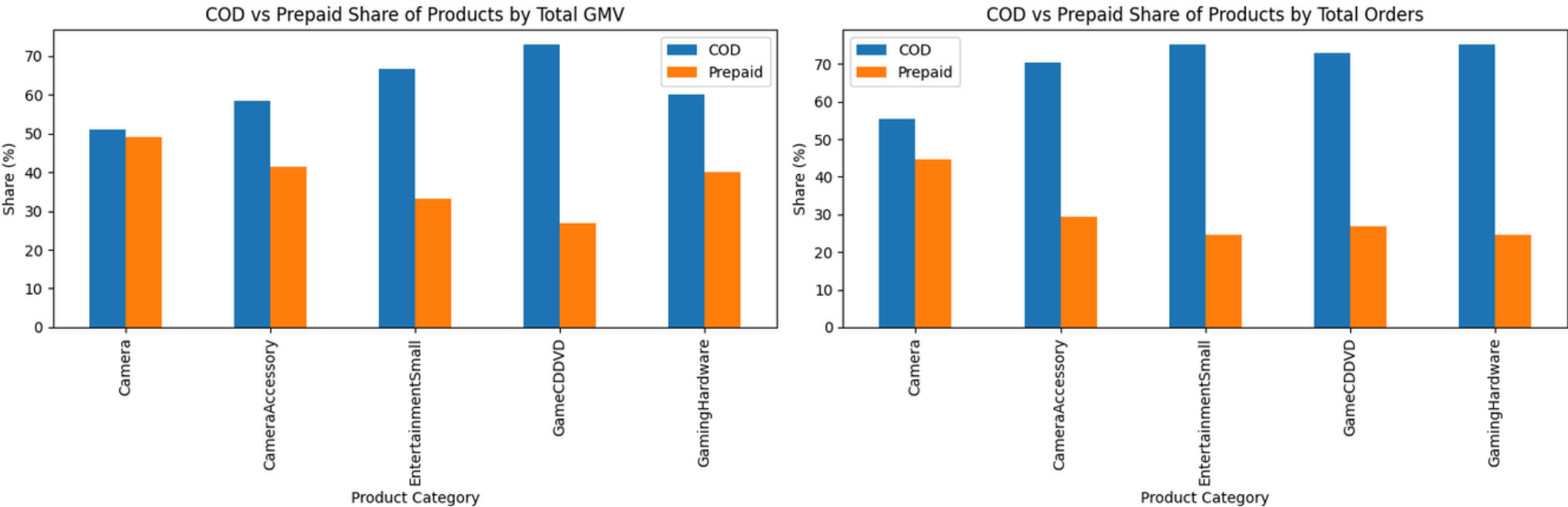


COD vs Pre-paid

COD drives volume, but prepaid brings higher value. While COD makes up 70%+ of sales, prepaid orders have higher value per transaction. Incentivize prepaid with cashback, EMI, and faster delivery to boost profitability and reduce return risks.



TV & Sponsorship focus on off-season brand-building, while Online and Digital ads peak during holidays for conversions. Content Marketing stays steady, ensuring consistent ROI. Prioritize high-ROI channels and cut low-impact spend for better budget efficiency.



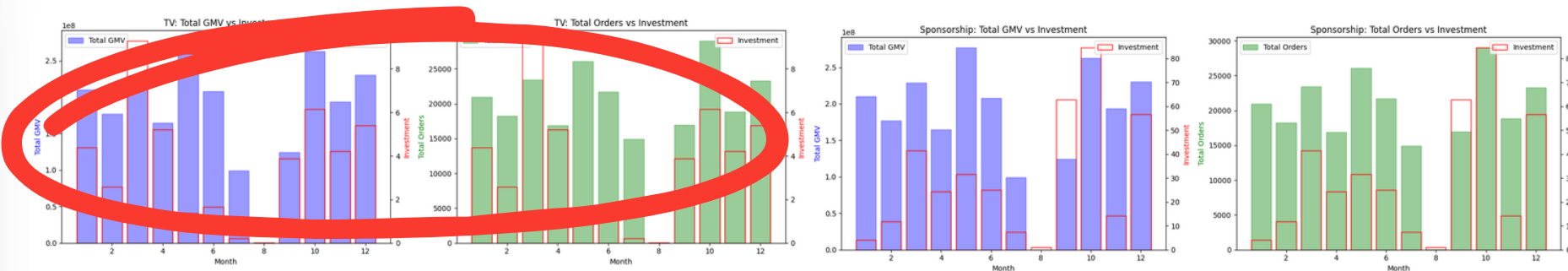
Box Plots, bar graphs, Correlation matrices

Measuring the Unmeasurable

Influence of Holidays & Payment Trends

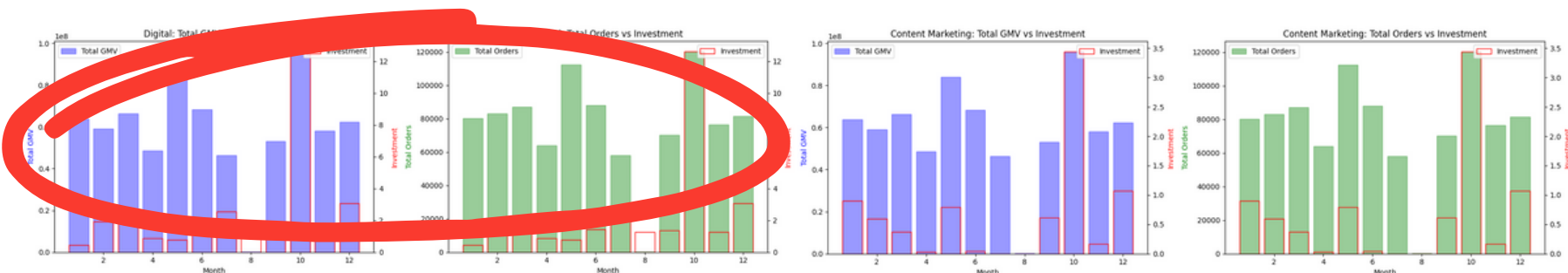
Luxury vs Mass

Total GMV vs Investment and Total GMV vs Orders for Luxury products

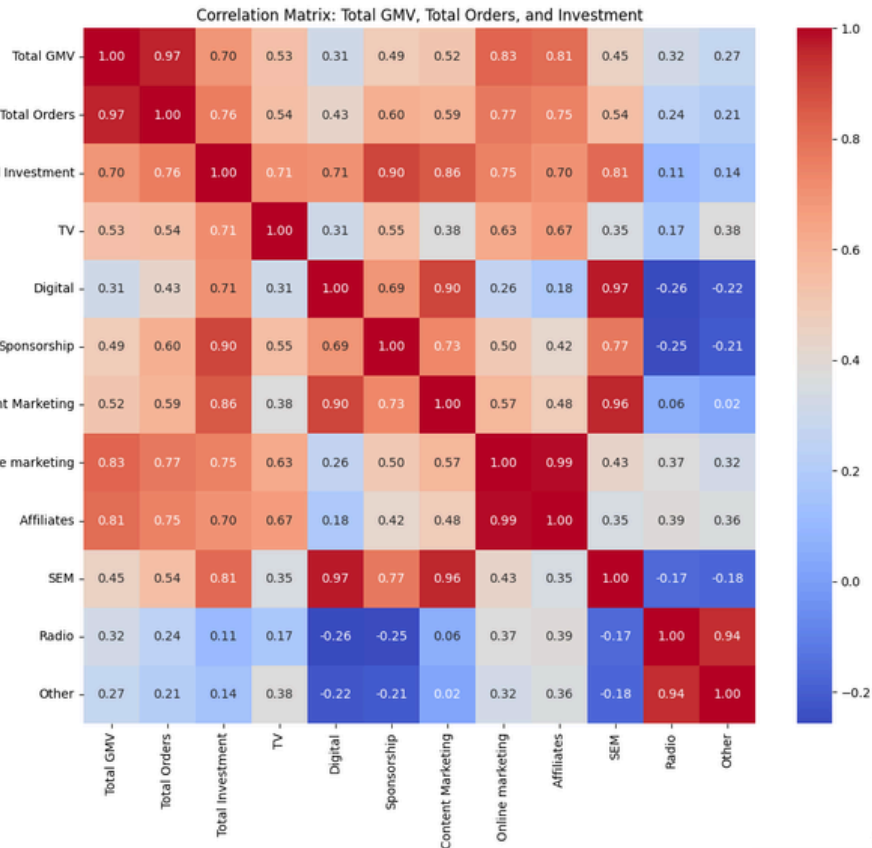


The analysis reveals that TV marketing investment does not consistently translate into higher GMV or order volume for luxury products, highlighting inefficiencies in spend allocation. While some months see a strong correlation between investment and returns, others (e.g., month 8) show disproportionate spending with minimal impact, signaling potential wastage. To maximize ROI, a strategic shift toward high-performing digital and premium-targeted marketing channels is crucial for luxury product success.

Total GMV vs Investment & Total GMV vs Orders for Mass Market products



The analysis shows that digital marketing investment strongly correlates with increased GMV and order volume for mass-market products, making it a high-impact channel. Unlike TV advertising, digital spend delivers consistent returns, especially in peak months like month 10, where a spike in investment directly drives record-high sales. To sustain and amplify growth, scaling digital investment strategically while optimizing spend efficiency can maximize revenue impact in the mass-market segment.

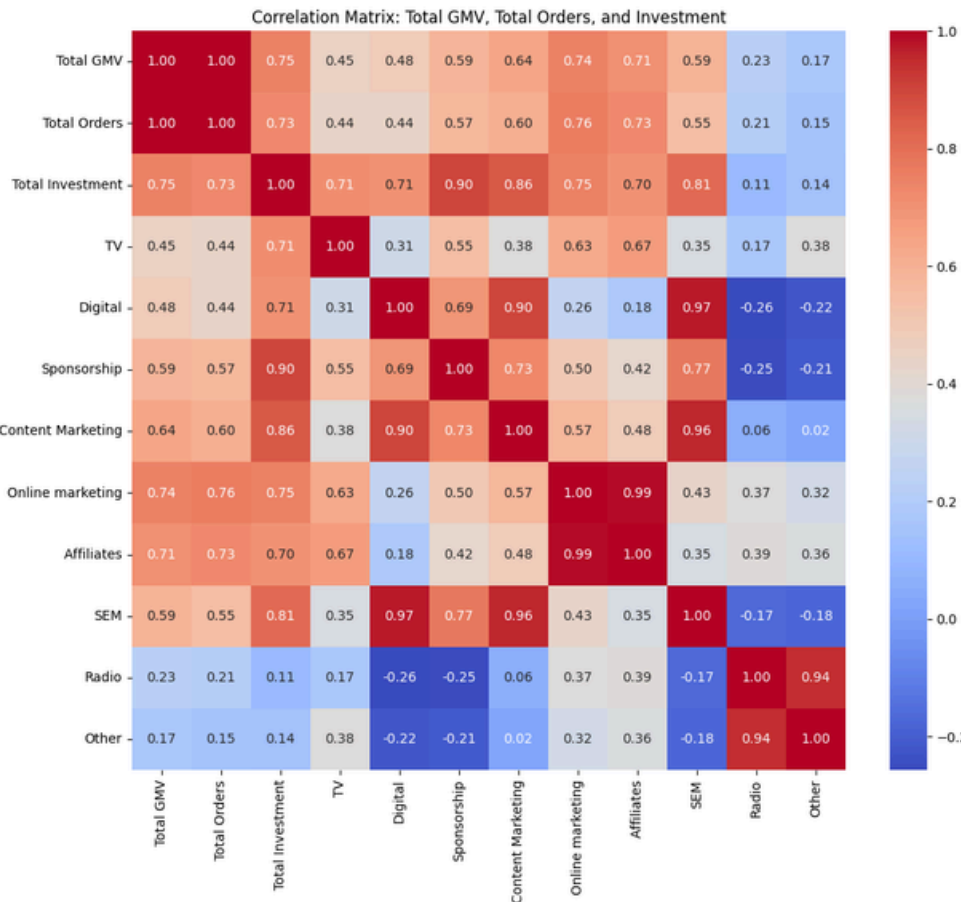


Luxury products

Online Marketing (0.83), Affiliates (0.81), and Total Investment (0.70+) drive high GMV, while TV (0.53) and Digital (0.31) have a weaker impact.

Mass Market products

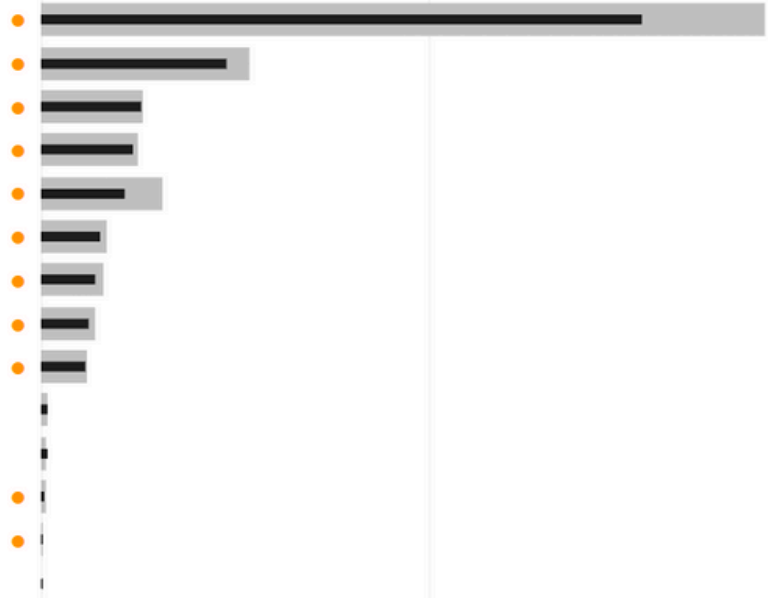
Total Investment (0.75), Online Marketing (0.74), and Content Marketing (0.64) strongly influence GMV, with Digital (0.48) performing better than in luxury.



DASHBOARD ON TABLEAU

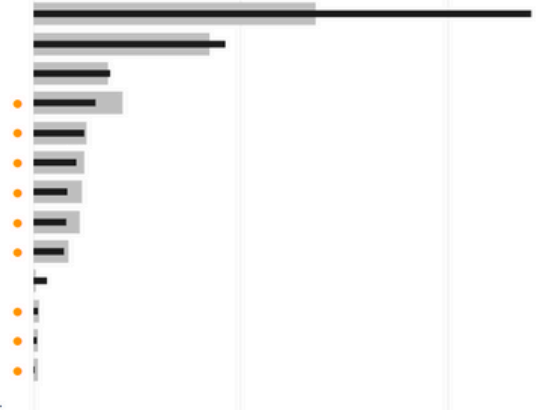
Sub-category Comparison for 2/2024

Camera
Speaker
HomeAudio
CameraAccessory
GamingConsole
Game
GamingAccessory
TVVideoSmall
AudioMP3Player
AudioAccessory
CameraStorage
HomeTheatre
AmplifierReceiver
GameMembership..



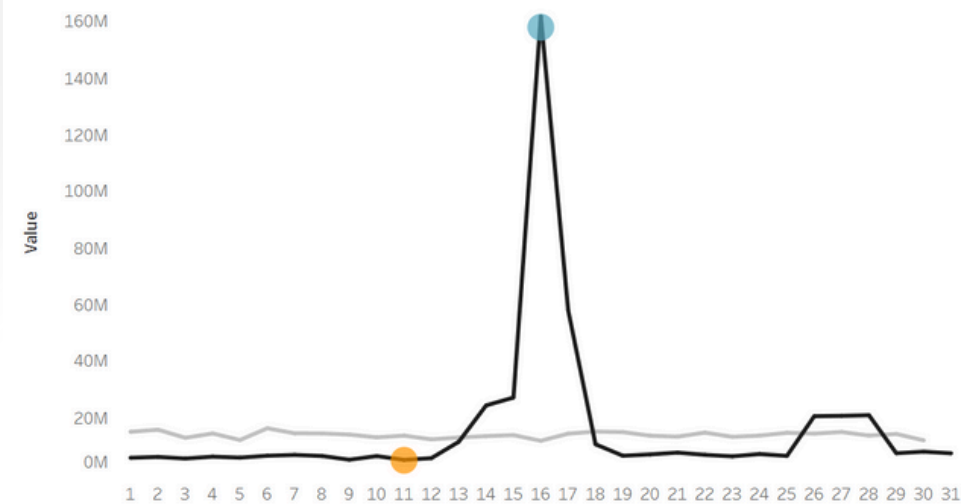
Sub-category Comparison for 10/2023

Camera
Speaker
HomeAudio
CameraAccessory
TVVideoSmall
Game
AudioMP3Player
GamingAccessory
GamingConsole
HomeTheatre
CameraStorage
AudioAccessory
AmplifierReceiver
GameMembership..



Total Revenue
₹503M

▲ 19.20% over last month



Select Year

2024

Select Month

2

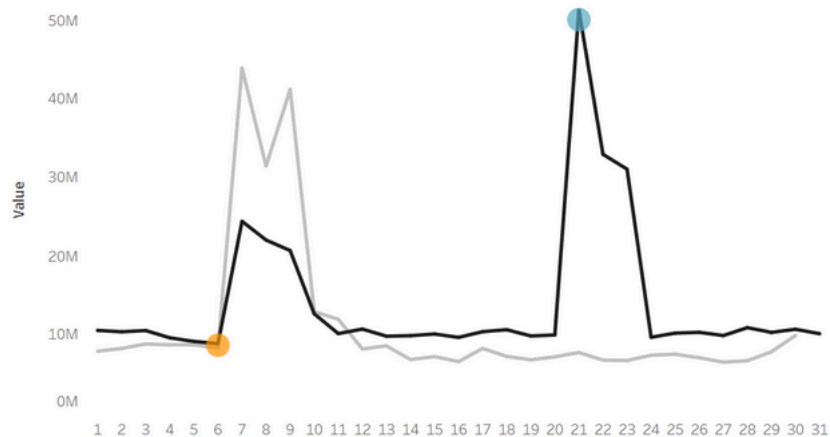
Count of CM Orders



Total Revenue

₹436M

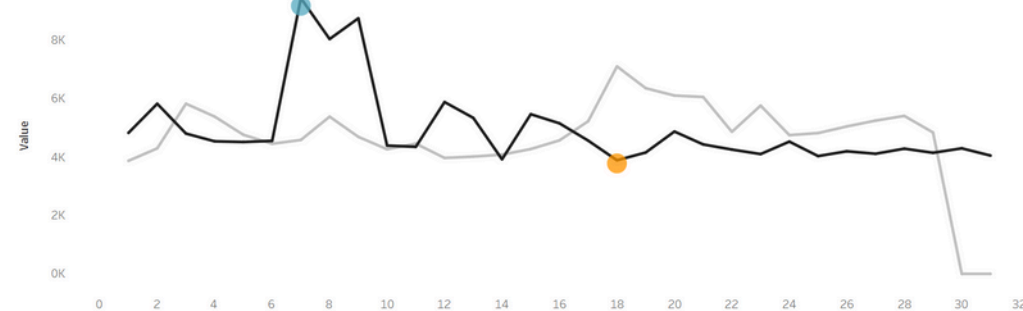
▲ 31.51% over last month



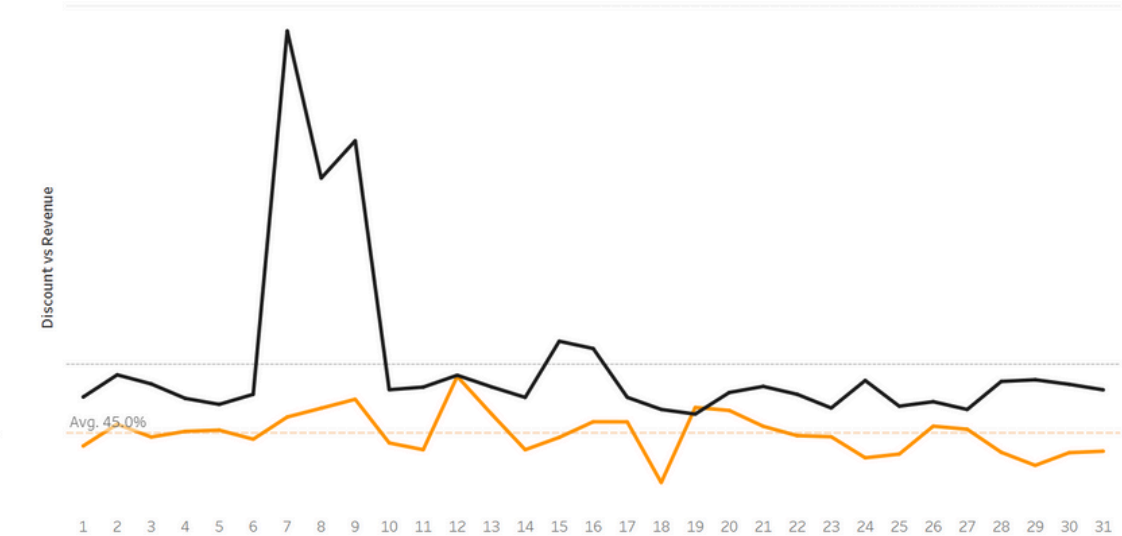
Total Orders

153K

▲ 7.13% over last month



Discount vs Revenue for 3-2024



Select Year

2023

Select Month

10

Measure Names

CM Revenue

PM Revenue

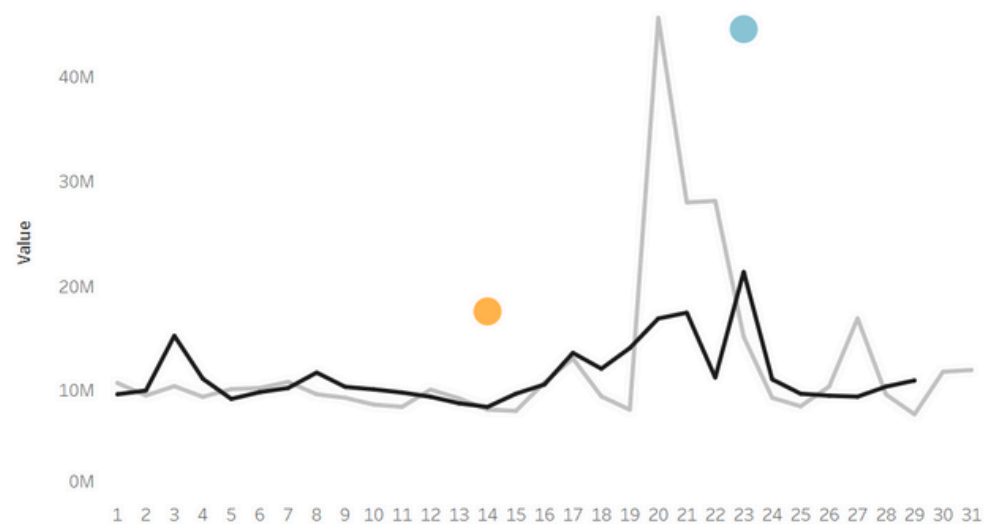
Min/Max Revenue



Total Revenue

₹332M

▼ -14.37% over last month



Select Year

2024

Select Month

2

Measure Names

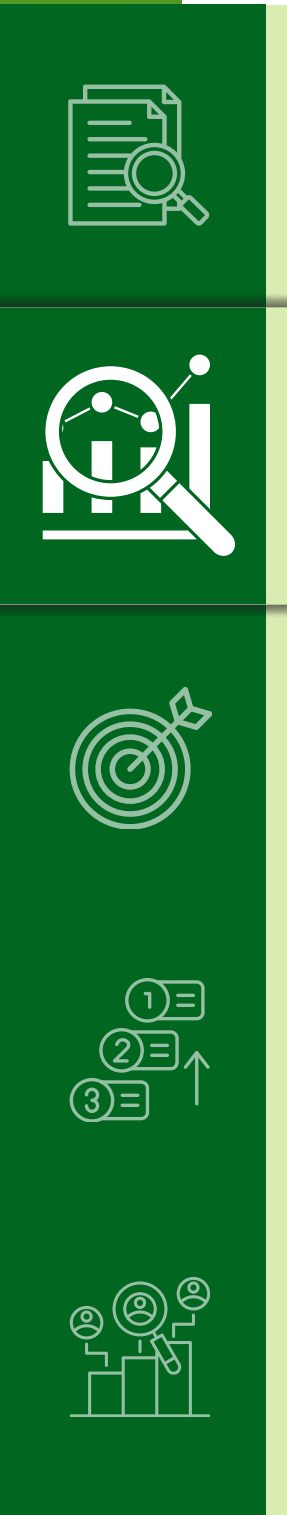
CM Revenue

PM Revenue

Min/Max Revenue

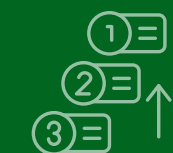


Key Performance Indicators (KPIs)



KPI	Strategy	Benchmark	Actionable Recommend-ations	Expected Business Impact	Potential Risks	Implementat-ion Time
Correlation b/w sales & marketing spends	Prioritize high-ROI channels (Online, Affiliates) and optimize TV & Digital spend.	Target ROI increase of 15% by shifting 20% budget to high-performing channels.	Reallocate budget towards digital and online channels while limiting offline inefficiencies.	Increased sales efficiency and improved ROI by 15-20%	Fluctuations in digital ad performance may impact marketing ROI	Short-term (0-3 months): Adjust budget allocation for high-ROI channels.
Product Positioning Matrix	Invest more in 'Stars' for growth, maintain 'Cash Cows' for profitability, and optimize 'Question Marks'	Increase revenue share of 'Stars' by 20% and reduce 'Dogs' share by 10%	Develop promotional strategies for 'Stars' and phase out 'Dogs' with low sales impact	Higher revenue from well-performing products and reduced losses from low-selling ones.	Improper categorization may lead to misallocation of resources.	Short-term (0-3 months): Realign investment strategies based on BCG matrix.
Top 3 Categories (GMV & Units Sold Trend)	Increase investment in the top 3 performing categories to drive revenue.	Achieve 15% YoY growth in top-performing categories.	Introduce bundle offers and targeted discounts in top-selling categories.	Boosted revenue by leveraging demand in strong-performing categories.	Demand shifts could impact performance of top categories	Mid-term (3-6 months): Optimize marketing for top-performing categories
Luxury vs. Mass Market	Luxury relies more on Online & Affiliates; Mass Market benefits from Digital & Content Marketing.	Optimize spend allocation to increase high-margin luxury sales by 10%.	Customize luxury campaigns for high-net-worth customers, focusing on digital and affiliate channels	Improved luxury product profitability and enhanced high-value customer retention	Market trends and customer behavior may vary, requiring continuous analysis.	Mid-term (3-6 months): Implement luxury-focused digital ad campaigns
Customer Acquisition Cost (CAC)	Optimize acquisition costs by balancing paid marketing with organic strategies	Maintain CAC below 10% of Average Order Value (AOV)	Balance digital marketing spend with organic lead generation strategies.	Lower acquisition costs and better long-term profitability	Unpredictable cost fluctuations may affect acquisition expenses	Mid-term (3-6 months): Introduce CAC monitoring and optimization strategies
Product MRP Distribution	Adjust pricing strategy based on MRP distribution to enhance revenue and market reach.	Ensure MRP alignment with competitor pricing for better market positioning.	Adjust pricing dynamically based on competitor analysis and demand forecasting.	Higher customer conversion rates due to competitive pricing strategies.	Pricing misalignment with competitors may reduce conversions.	Short-term (0-3 months): Align pricing with market trends.
COD vs Prepaid GMV Share	Encourage prepaid payments with discounts and reduce COD-related costs.	Increase prepaid transactions by 10% to improve cash flow and reduce logistics costs.	Offer exclusive benefits for prepaid transactions to encourage adoption.	Reduced COD risks and improved cash flow management.	Customer preference for COD may persist despite incentives.	Short-term (0-3 months): Implement prepaid discount strategies.
Correlation of Units Sold Across Products	Optimize inventory and pricing by understanding demand correlations among products.	Identify at least 3 strong product correlations for better inventory planning.	Use correlation insights for inventory optimization and targeted marketing.	Better stock management and improved marketing efficiency.	Seasonality and external factors may affect correlation patterns.	Short-term (0-3 months): Improve stock planning using correlation analysis.

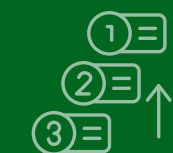
Key Performance Indicators (KPIs)



KPI	Strategy	Benchmark	Actionable Recommendations	Expected Business Impact	Potential Risks	Implementat-ion Time
Ad Spend by Month	Shift ad spend to high-performing months and reduce inefficient spending periods.	Improve ad spend efficiency by cutting underperforming months by 15%.	Monitor seasonal trends and optimize budget allocation based on past ad performance.	Reduced ad waste and increased returns on marketing investment.	Unforeseen market trends may require flexible ad strategies.	Short-term (0-3 months): Adjust ad spend based on historical trends.
Ad Spend on Holiday vs. Non-Holiday	Increase holiday-specific promotions for better seasonal conversions.	Increase holiday-season GMV by 20% through optimized marketing campaigns.	Launch aggressive promotions during peak shopping seasons to maximize conversion rates.	Better holiday season sales and improved customer targeting.	Holiday promotions may face high competition from other retailers.	Mid-term (3-6 months): Execute holiday-focused campaigns.
Sales in Different Temperature Ranges	Target warm-weather sales for high-performing products and plan discounts for snowfall periods.	Align sales peaks with optimal weather conditions to maximize revenue.	Develop targeted pricing strategies for different weather conditions.	Higher sales in optimal weather conditions and seasonal adjustments.	Weather unpredictability could affect forecasting accuracy.	Long-term (6+ months): Implement weather-based demand strategies.
NPS Score	Improve customer experience through targeted engagement strategies based on NPS insights.	Increase NPS score to above 50 for strong customer satisfaction.	Implement customer feedback systems and loyalty programs to improve satisfaction.	Stronger customer loyalty and higher retention rates.	Negative feedback loops could impact brand reputation.	Long-term (6+ months): Establish continuous customer satisfaction improvements.
Bivariate Analysis: Sales vs. Orders & Spending	Align marketing spend with sales trends to ensure effective budget allocation.	Ensure a minimum 0.7 correlation between marketing spend and sales trends.	Reallocate marketing spend based on high-conversion months and seasonal demand.	Optimized marketing investments with better alignment to sales performance.	Changes in consumer behavior could alter previous trends.	Mid-term (3-6 months): Refine marketing spending patterns.

Decoding ElectroMart

Impact Analysis on Marketing ROI



Marketing Channel	Spend (₹ Cr)	ROI (ratio)	CAC (₹)	Budget Action
TV	44.368068	5.501802	468.104160	Maintain Budget
Digital	29.660227	8.725893	312.929694	Increase Budget
Sponsorship	365.367769	-0.210461	3854.806047	Reallocate/Eliminate
Content Marketing	8.027973	34.933377	84.698989	Increase Budget
Online Marketing	193.639614	0.489737	2042.991247	Reallocate/Eliminate
Affiliates	61.36784	3.700557	647.480801	Reduce Budget
SEM	91.202143	2.162998	962.226565	Reduce Budget
Radio	4.670000	60.771348	49.270751	Increase Budget
Other	48.022545	5.007016	506.660995	Maintain Budget

- 1

Sales Peaks During Discounts

Discounts boost GMV by increasing customer traffic, leading to higher sales volumes that outweigh margin reductions, ultimately driving overall revenue growth.
- 2

Holidays vs. Paydays on Sales

Holidays boost sales, while paydays have minimal impact, making holiday promotions more effective.
- 3

High ROI Channels

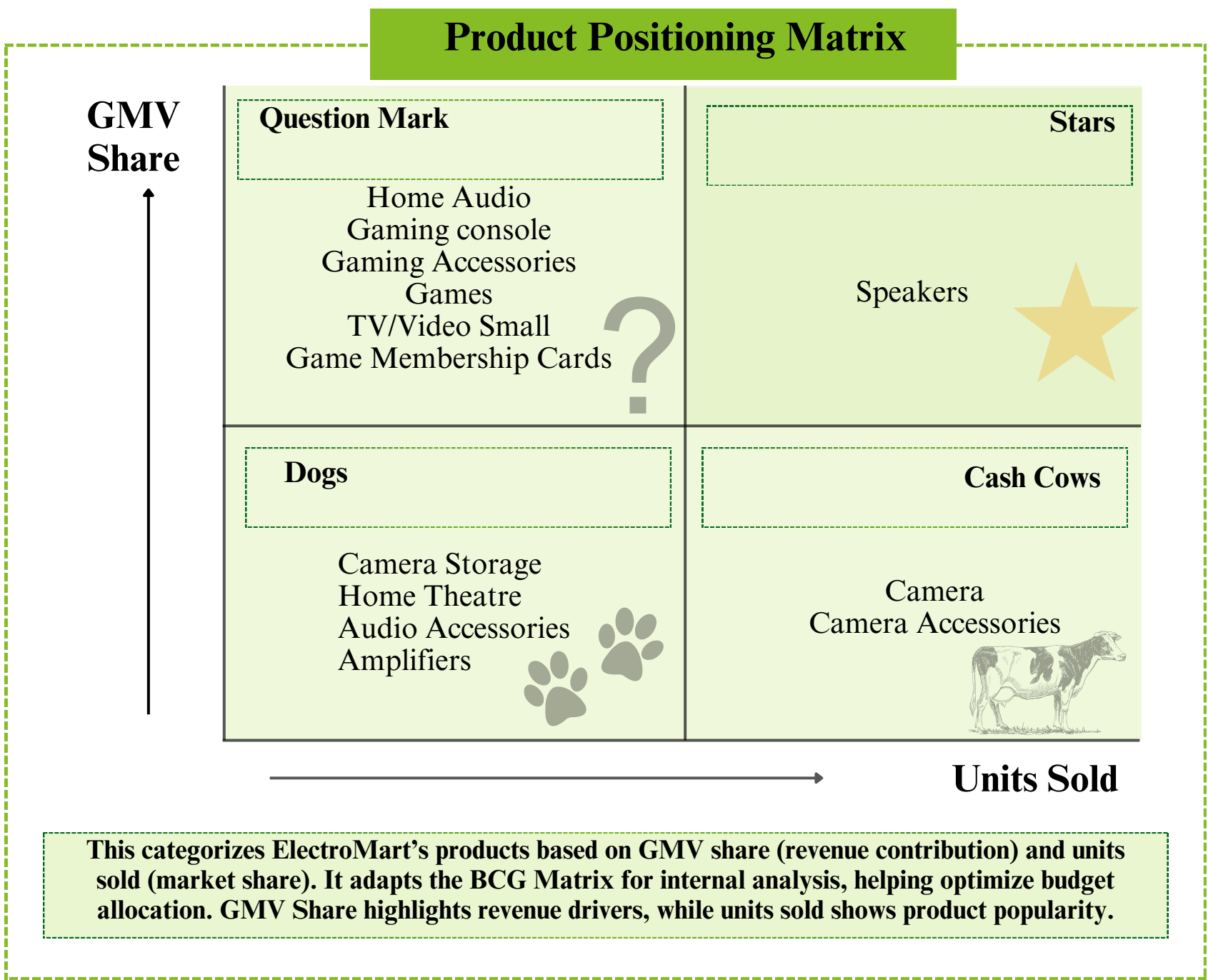
Content Marketing has the highest ROI, while Digital & TV ads also deliver strong revenue returns.
- 4

Budget Allocation

Sponsorships consume 43.17% of the budget but show low ROI, while Online Marketing (22.88%) has moderate efficiency, needing optimization.
- 4

Underutilized Channels

Radio & Other channels show high ROI despite low budget allocation, indicating potential missed opportunities or data inconsistencies.



- ### Terminology

 - Stars** – High revenue & high sales; invest for growth.
 - Cash Cows** – High revenue & low sales; maintain for steady profit.
 - Question Marks** – Low revenue & high sales; evaluate for potential growth.
 - Dogs** – Low revenue & low sales; discontinue or reposition.

CFO's Problem Solved

Statistical Analysis and Budget Allocation

Different models with R2 scores and MSE

Models	R2 Score	MSE	Insights
Decision Tree	0.9962	102084.9612	It has highest R2 Score and lowest error
AdaBoost	0.8743	3883002.955	Low R2 score suggests that it might struggle with capturing relationships
Linear Regression	0.8486	4074827.72	Similar to Multiple Linear Regression, providing a solid baseline
Lasso Regression	0.8942	2848980.57	Improves generalization by reducing overfitting, making it more stable for moderately complex datasets.
XGBoost	0.9825	470731.803	High R2 score, has both high accuracy and stability on large data

BEST MODEL : Decision Tree

R² score is 0.9962, indicating an excellent fit to the data, and one of the lowest Mean Squared Errors (MSE = 102084.9612), suggesting that its predictions are highly accurate.

Promotional Strategy

Promotion Type	Impact	Strategy
Holiday Promotions	Significant increase in GMV and customer engagement	Increase holiday season budget for digital ads and online campaigns
Discount-Driven Sales	Discounted products see spikes in sales, proving their effectiveness	Continue seasonal discount promotions with strategic stock planning
Ad Spend on Digital & TV	Digital Ads show immediate impact, TV Ads contribute to long-term brand awareness	Allocate more towards Digital for immediate conversions and TV for awareness
Radio & Content Marketing	High ROI but low budget allocation	Increase investment moderately to test further impact
Sponsorship Budget Reallocation	Sponsorships consume too much budget but don't yield proportional sales impact	Reduce sponsorship spending and focus on direct digital conversion channels

Product Specific Recommendations

Product Category	Marketing Channel Effectiveness	Strategic Recommendation
Luxury Products(High margin, Low Volume)	Digital Ads, Content Marketing, and Influencer Sponsorships work best	Focus on premium branding, influencer campaigns, and financing options
Mass-Market Products (Low margin, High volume)	Online Ads & Discount Campaigns drive higher sales	Implement aggressive discount campaigns and performance marketing strategies

Budget Optimization Strategy

Ad Channels	Budget allocated (Rs)
TV	147822.137117
Digital	442495.557817
Sponsorship	400099.431852
Content Marketing	43039.124251
Online Marketing	183469.001652
Affiliates	24561.326797
SEM	837842.668157
Radio	899999.999999
Other	3096.562743

Maximun GMV (estimated) per order= 5993.40252036595

Monthly revenue generated = 52.1689 Cr

THANK YOU